

Motivation (the big picture). For a complex stochastic system such as a humanoid robot, we want to formally prove safety—for example, that it will not fall while walking.

Motivation (the realistic picture). Previous work [1] assumes that V is twice continuously differentiable. Weakening this assumption could lead to faster verification.

Introduction

We study stochastic differential equations

$$dX_t = f(X_t) dt + \sigma(X_t) dW_t, \quad X_t \in K \subset \mathbb{R}^d,$$

on compact K . Given initial, target, and unsafe sets X_0, X_T, X_u , choose $0 \leq \alpha < \beta$ and a *nonnegative* $V : K \rightarrow \mathbb{R}$ satisfying

$$\sup_{X_0} V \leq \alpha, \quad \inf_{X_u} V \geq \beta, \quad \inf_{\partial K \setminus X_T} V \geq \beta.$$

Let τ be the stopping time of reaching X_T , ∂K , or $\{V \geq \beta\}$. This key theorem, as a consequence of Ville’s inequality, connects the construction of a supermartingale to safety bounds for the SDE.

Theorem (reach–avoid guarantee). If $V(X_{t \wedge \tau})$ is a supermartingale, then

$$\mathbb{P}(\tau_{X_u} \wedge \tau_{\partial K} < \tau_{X_T}) \leq \alpha/\beta.$$

So the probability of reaching the unsafe set or leaving the domain before reaching the target set is bounded by α/β .

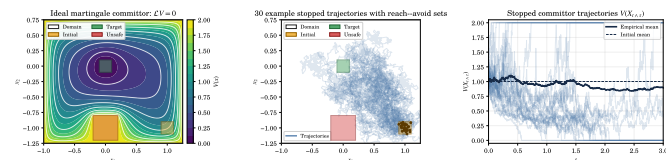


Figure 1. An ideal reach–avoid certificate and realizations of the stopped martingale $V(X_{t \wedge \tau})$. It’s numerically estimated to be a martingale.

Following the motivations given above, these research questions arise:

Research Question 1: Can non- C^2 certificates be valid?

Finding: Yes. By applying the Itô–Tanaka–Meyer formula we notice that we have to additionally account for local time.

Research Question 2: What is a useful class of certificates?

Finding: We identify three important properties: piecewise-quadratic structure, representational power, and local-time safety by construction through our **Deep Residual ICNN**.

Research Question 3: Do these certificates work in practice?

Finding: Yes, but so far only in very simplified settings. We identify useful techniques and several failure modes.

1. Validity of non- C^2 certificates

Let V be continuous and C^2 on each polyhedral cell K_i . Across a shared facet F , we have a normal vector $n_{i \rightarrow j, F} \in \mathbb{R}^d$ from K_i to K_j . Define the normal-gradient jump to be

$$\gamma_F(x) = (\nabla V_j(x) - \nabla V_i(x))^\top n_{i \rightarrow j, F}.$$

The multidimensional Itô–Tanaka formula gives

$$V(X_{t \wedge \tau}) = V(X_0) + \int_0^{t \wedge \tau} \mathcal{L}V(X_s) ds + M_t + \frac{1}{2} \sum_F \int_0^{t \wedge \tau} \gamma_F(X_s) dL_s^F,$$

where

$$\mathcal{L}V = \nabla V^\top f + \frac{1}{2} \text{tr}(\sigma \sigma^\top \nabla^2 V),$$

and L^F is nondecreasing surface local time.

Theorem (Itô–Tanaka certificate). Assume the set constraints above, the stochastic integral M_t is a true martingale, and below level β outside X_T :

$$\mathcal{L}V_i(x) \leq 0, \quad \gamma_F(x) \leq 0$$

on every cell and every shared facet. Then $V(X_{t \wedge \tau})$ is a supermartingale and the reach–avoid bound holds.

Proof sketch. For $s \leq t$, apply the formula on $(s, t]$ and condition on $\mathcal{F}_s$. The generator integral is nonpositive; $dL^F \geq 0$ and $\gamma_F \leq 0$ make every interface integral nonpositive; the martingale increment has conditional mean zero. Hence

$$\mathbb{E}[V(X_{t \wedge \tau}) | \mathcal{F}_s] \leq V(X_{s \wedge \tau}). \quad \square$$

2. A useful class of non- C^2 certificates

2.1 Piecewise-quadratic structure

We propose that V has this form:

$$V(x) = \sum_{k=1}^m \lambda_k \phi(z_k(x)) + c, \quad z(x) = \text{ReLU}(x),$$

where ϕ is a fixed continuous piecewise-quadratic activation. The ReLU network partitions K into polyhedral cells and, after refinement by ϕ ,

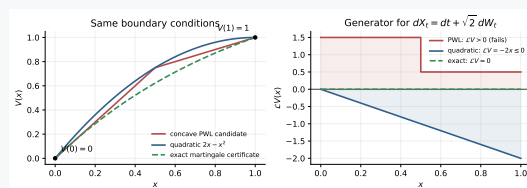
$$V_i(x) = c_i + p_i^\top x + \frac{1}{2} x^\top Q_i x.$$

Thus $\nabla V_i = p_i + Q_i x$ and $\nabla^2 V_i = Q_i$, which we apply when checking the generator and local-time conditions.

Why is plain ReLU insufficient? For $dX_t = dt + \sqrt{2} dW_t$ on $[0, 1]$, suppose $V(0) < V(1)$. A continuous piecewise-linear V satisfying $\gamma_F \leq 0$ has nonincreasing slopes, but must begin with some positive slope a_i . Inside that piece,

$$\mathcal{L}V = V' + V'' = a_i > 0,$$

contradicting the generator condition. On the other hand, the quadratic $V(x) = 2x - x^2$ works: $\mathcal{L}V - 2x \leq 0$.



2.2 Representational power

Complex SDEs can require many quadratic regions. Deep neural networks can represent these regions compactly and discover their partition during training. Piecewise quadratics also have strong approximation power: under strict margins, a smooth 2D certificate can be approximated by valid C^1 piecewise-quadratic splines. Following existing committor-function literature, which uses deep neural networks to train committors [3,4], we use deep networks to learn certificate-like functions.

2.3 Local-time safety by construction

Checking the local-time condition creates a verification bottleneck. If the network induces N full-dimensional regions, naively testing all pairs costs $O(N^2)$, even though only adjacent regions sharing a facet can contribute local time. Explicit adjacency discovery is already expensive as depth and width increase.

To guarantee the sign without discovering every facet, we propose the **Deep Residual ICNN** architecture

$$V_\theta(x) = \rho(S_\theta(x) - C_\psi(x)), \quad \rho > 0.$$

The C^1 piecewise-quadratic branch may be

$$S_\theta(x) = c + p^\top x + \frac{1}{2} x^\top H x + \frac{1}{2} \sum_k v_k \text{ReLU}(a_k^\top x + b_k)^2.$$

The convex piecewise-affine branch is a deep ReLU ICNN [2]:

$$\begin{aligned} z_1 &= \text{ReLU}(U_1 x + b_1), \\ z_\ell &= \text{ReLU}(U_\ell x + b_\ell + W_\ell z_{\ell-1}), \\ C_\psi(x) &= u^\top x + c + w^\top z_L, \quad W_\ell, w \geq 0. \end{aligned}$$

Nonnegative hidden-to-hidden and output weights make C_ψ convex; ReLU makes it continuous and piecewise affine. Soft-plus parameterization enforces the weight constraints during unconstrained training.

Theorem (local-time condition by construction). For every pair of adjacent cells and every parameter choice,

$$(\nabla V_{\theta, j} - \nabla V_{\theta, i})^\top n_{i \rightarrow j, F} \leq 0.$$

Therefore all codimension-one local-time contributions are nonpositive; no activation-region enumeration is required to establish their sign.

Proof sketch for §2.3. Restrict convex C_ψ to a line normal to F . The derivative of a one-dimensional convex function is nondecreasing, so

$$(\nabla C_{\psi, j} - \nabla C_{\psi, i})^\top n_{i \rightarrow j, F} \geq 0.$$

Because $S_\theta \in C^1$, its gradient jump is zero. Consequently

$$(\nabla V_{\theta, j} - \nabla V_{\theta, i})^\top n = -\rho(\nabla C_{\psi, j} - \nabla C_{\psi, i})^\top n \leq 0. \quad \square$$

This controls only interface curvature. Nonnegativity, set separation, the interior generator inequality, and integrability must still be verified.

3. Experiments and failure modes

At first, we tried to train a plain ReLU MLP with a piecewise-quadratic output layer and found that it was very difficult to train.

After proposing the Deep Residual ICNN architecture, training became easier and we found valid certificates for a simple SDE with loose bounds.

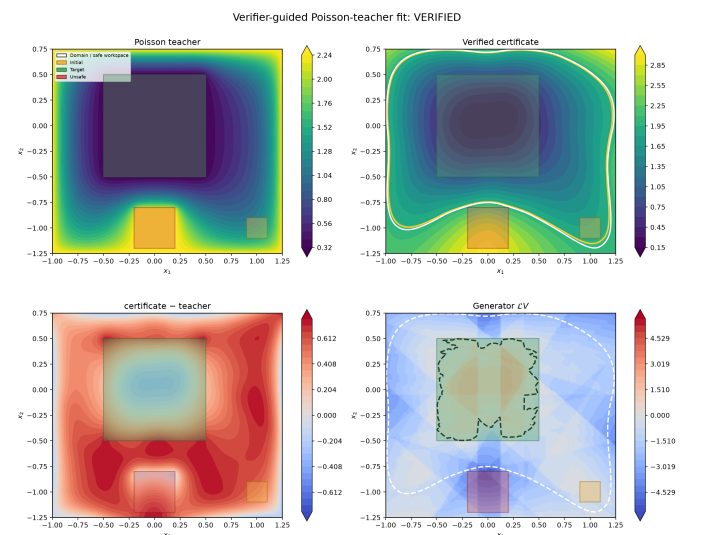


Figure 3. A verifier-guided Poisson-teacher fit: the teacher, verified certificate, approximation error, and verified generator.

Limitations and future work

1. We could not train a valid certificate using Adam alone; success required linear programming on the S_{θ_1} branch.
2. The ICNN branch C_{θ_2} was almost zero and contributed little to the final certificate.

We found a connection to a large body of literature on committor functions, which offers efficient methods for training certificate-like functions [3,4]. Its sampling methods improved our training with Adam, although they have not yet produced a verified certificate.

These results remain limited to simplified settings. The main open challenge is to make both branches train effectively while retaining exact verification.

Preliminaries and notation

Martingales. On a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$, an adapted integrable process Y_t is a supermartingale when

$$\mathbb{E}[Y_t | \mathcal{F}_s] \leq Y_s, \quad s \leq t.$$

It is a martingale when equality holds.

Generator. For the SDE above,

$$\mathcal{L}V = \nabla V^\top f + \frac{1}{2} \text{tr}(\sigma \sigma^\top \nabla^2 V).$$

The first term is the Lie derivative along the deterministic drift; the second records curvature sampled by the diffusion.

Local time. L_t^F measures the infinitesimal time accumulated by X_t at a kink facet F . Its contribution is weighted by the signed normal-gradient jump γ_F .

Python package

We propose a Python package, `tanaka_certificates`, with utilities for verifying Tanaka certificates and a thorough test suite. Example usage:

```
1 # pip install tanaka-certificates
2
3 sde, problem = make_piecewise_quadratic_ou_2d_problem()
4 verifier = VerifierPiecewiseQuadratic(
5     sde, problem, _ou_example_certificate())
6
7 assert verifier.verify() is VerificationResult.NOT_VERIFIED
8 issue = next(i for i in verifier.issues
9             if i.kind is IssueKind.DOMAIN_BOUNDARY)
10 assert issue.value < problem.beta
```

Listing 1: Verify a piecewise-quadratic certificate.